TALLER –FIREWALL PERSONAL CON IPTABLES

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En equipos conformados por 3 estudiantes, instalar Linux mediante imagen en una máquina virtual de virtual box y realizar un script en Shell para diseñar un firewall personal que contenga además las restricciones dadas.

- ✓ Permitir ssh.
- ✓ Política por defecto.
- ✓ Permitir tráfico para la interfaz loopback.
- ✓ Peticiones y respuestas protocolo ICMP.
- ✓ Consultas y respuestas DNS.
- ✓ Tráfico http y https utilizando la extensión multiport.
- ✓ Deniega el acceso a tu servidor web desde una ip concreta.
- ✓ No permitir el acceso al servidor web de www.josedomingo.org (Tienes que utilizar la ip, con nslookup puedes averiguarla).
- ✓ Permitir los accesos solo desde la ip de un cliente específico al servidor MySql del equipo. Comprueba que desde otro cliente no se puede acceder.

Entregar el script de Shell y pantallazos de la práctica.

Desarrollo

```
sebastian@UbuntuPc:~$ sudo apt update
[sudo] password for sebastian:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,542 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [977 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.8 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,842 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,825 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [655 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,569 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [320 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.1 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7,276 B]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,340 B]
Get:27 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
```

- Muestra la configuración inicial de la máquina virtual con Linux. Sirve para comprobar que el sistema ya está instalado correctamente y listo para comenzar el taller del firewall.

```
  GNU nano 7.2                    /etc/firewall.sh

                                 [ New File ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify   ^/ Go To Line
```

- Es una terminal de Linux abierta.
  Aquí se verifica que el sistema funciona y que se pueden ejecutar comandos (esto es necesario porque el firewall se hace con un script en shell).

```
  GNU nano 7.2                                    /etc/firewall.sh *

IFACE="enp0s3"
IP_BLOQUEADA="10.0.2.15"
IP_JOSEDOMINGO="127.0.0.53#53"
IP_CLIENTE_MYSQL="10.0.2.15"


RED='\033[0;31m'; GREEN='\033[0;32m'; YELLOW='\033[1;33m'; NC='\033[0m'
info()  { echo -e "${GREEN}[+]${NC} $*"; }
warn()  { echo -e "${YELLOW}[!]${NC} $*"; }
error() { echo -e "${RED}[-]${NC} $*"; }

[[ $EUID -ne 0 ]] && { error "Ejecuta como root (sudo)."; exit 1; }

limpiar() {
    info "Limpiando reglas..."
    iptables -F; iptables -X; iptables -Z
    iptables -t nat -F; iptables -t mangle -F
}

iniciar() {
    limpiar
```

- Se muestra el uso del comando iptables por primera vez.
  La imagen sirve para comprobar que el firewall está activo y que se pueden ver las reglas actuales.

```
    # 2. Loopback
    info "Loopback..."
    iptables -A INPUT  -i lo -j ACCEPT
    iptables -A OUTPUT -o lo -j ACCEPT

    # 3. Conexiones establecidas / relacionadas
    info "ESTABLISHED/RELATED..."
    iptables -A INPUT  -m state --state ESTABLISHED,RELATED -j ACCEPT
    iptables -A OUTPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
    iptables -A INPUT  -m state --state INVALID -j DROP
    iptables -A OUTPUT -m state --state INVALID -j DROP

    # 4. SSH
    info "SSH (22)..."
    iptables -A INPUT  -p tcp --dport 22 -m state --state NEW,ESTABLISHED -j ACCEPT
    iptables -A OUTPUT -p tcp --sport 22 -m state --state ESTABLISHED     -j ACCEPT

    # 5. ICMP
    info "ICMP ping/pong..."
    iptables -A INPUT  -p icmp --icmp-type echo-request -j ACCEPT
    iptables -A OUTPUT -p icmp --icmp-type echo-reply   -j ACCEPT
    iptables -A OUTPUT -p icmp --icmp-type echo-request -j ACCEPT
    iptables -A INPUT  -p icmp --icmp-type echo-reply   -j ACCEPT

    # 6. DNS
    info "DNS (53 UDP/TCP)..."
    iptables -A OUTPUT -p udp --dport 53 -j ACCEPT
```

- Se ejecuta un comando como iptables -L. La imagen sirve para **ver las reglas actuales del firewall** antes de empezar a configurarlo.

```
# 6. DNS
info "DNS (53 UDP/TCP)..."
iptables -A OUTPUT -p udp --dport 53 -j ACCEPT
iptables -A INPUT  -p udp --sport 53 -j ACCEPT
iptables -A OUTPUT -p tcp --dport 53 -j ACCEPT
iptables -A INPUT  -p tcp --sport 53 -j ACCEPT

# 7. HTTP/HTTPS con multiport
info "HTTP/HTTPS multiport (80,443)..."
iptables -A OUTPUT -p tcp -m multiport --dports 80,443 -j ACCEPT
iptables -A INPUT  -p tcp -m multiport --dports 80,443 -j ACCEPT

# 8. Bloquear IP concreta al servidor web
warn "Bloqueando ${IP_BLOQUEADA} -> web..."
iptables -A INPUT -p tcp -m multiport --dports 80,443 \
         -s "${IP_BLOQUEADA}" -j DROP

# 9. Bloquear josedomingo.org
warn "Bloqueando josedomingo.org (${IP_JOSEDOMINGO})..."
iptables -A INPUT  -p tcp -m multiport --dports 80,443 -s "${IP_JOSEDOMINGO}" -j DROP
iptables -A OUTPUT -p tcp -m multiport --dports 80,443 -d "${IP_JOSEDOMINGO}" -j DROP

# 10. MySQL solo desde cliente autorizado
info "MySQL solo para ${IP_CLIENTE_MYSQL}..."
iptables -A INPUT -p tcp --dport 3306 ! -s "${IP_CLIENTE_MYSQL}" -j DROP
iptables -A INPUT  -p tcp --dport 3306 -s "${IP_CLIENTE_MYSQL}" \
         -m state --state NEW,ESTABLISHED -j ACCEPT
```

- Aquí se configura la política por defecto (default policy). Significa que el sistema bloqueará todo el tráfico si no hay una regla que lo permita.

```
sebastian@UbuntuPc:~$ nslookup www.josedomingo.org
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
www.josedomingo.org     canonical name = endor.josedomingo.org.
Name:   endor.josedomingo.org
Address: 37.187.119.60

sebastian@UbuntuPc:~$
```

- Se muestra una regla para permitir el tráfico de la interfaz loopback (lo). Esto es necesario para que el sistema funcione correctamente internamente.

```
Fetched 11.2 MB in 4s (3,087 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
39 packages can be upgraded. Run 'apt list --upgradable' to see them.
sebastian@UbuntuPc:~$ sudo apt install iptables-y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package iptables-y
sebastian@UbuntuPc:~$ sudo nano /etc/firewall.sh
```

- Se ve una regla que permite ICMP (ping). Sirve para que el equipo pueda enviar y recibir pings sin que el firewall lo bloquee.

```
sebastian@UbuntuPc:~$ sudo chmod +x /etc/firewall.sh
[sudo] password for sebastian:
sebastian@UbuntuPc:~$
```

- Aparece una regla que permite conexiones SSH (puerto 22). Esto permite conectarse remotamente al sistema sin problemas.

```
sebastian@UbuntuPc:~$ sudo /etc/firewall.sh start
/etc/firewall.sh: 13: [[: not found
-e [+] Limpiando reglas...
-e [+] Política DROP en INPUT, OUTPUT, FORWARD...
-e [+] Loopback...
-e [+] ESTABLISHED/RELATED...
-e [+] SSH (22)...
-e [+] ICMP ping/pong...
-e [+] DNS (53 UDP/TCP)...
-e [+] HTTP/HTTPS multiport (80,443)...
-e [!] Bloqueando 10.0.2.15 -> web...
-e [!] Bloqueando josedomingo.org (127.0.0.53#53)...
iptables v1.8.10 (nf_tables): host/network `127.0.0.53#53' not found
Try `iptables -h' or 'iptables --help' for more information.
iptables v1.8.10 (nf_tables): host/network `127.0.0.53#53' not found
Try `iptables -h' or 'iptables --help' for more information.
-e [+] MySQL solo para 10.0.2.15...
-e [+] === Firewall activo ===
sebastian@UbuntuPc:~$
```

- Se observa una regla para permitir consultas DNS (puerto 53). Esto permite que el sistema pueda navegar por internet.

```
sebastian@UbuntuPc:~$ sudo /etc/firewall.sh status
/etc/firewall.sh: 13: [[: not found

=== INPUT ===
Chain INPUT (policy DROP 2 packets, 244 bytes)
num   pkts bytes target     prot opt in     out     source               destination
1       16  1600 ACCEPT     0    --  lo     *       0.0.0.0/0            0.0.0.0/0
2      198 24891 ACCEPT     0    --  *      *       0.0.0.0/0            0.0.0.0/0            state RELATED,ESTABLISHED
3        0     0 DROP       0    --  *      *       0.0.0.0/0            0.0.0.0/0            state INVALID
4        0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:22 state NEW,ESTAB
LISHED
5        0     0 ACCEPT     1    --  *      *       0.0.0.0/0            0.0.0.0/0            icmptype 8
6        0     0 ACCEPT     1    --  *      *       0.0.0.0/0            0.0.0.0/0            icmptype 0
7        0     0 ACCEPT     17   --  *      *       0.0.0.0/0            0.0.0.0/0            udp spt:53
8        0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:53
9        0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            multiport dports 80,443
10       0     0 DROP       6    --  *      *       10.0.2.15            0.0.0.0/0            multiport dports 80,443
11       0     0 DROP       6    --  *      *      !10.0.2.15            0.0.0.0/0            tcp dpt:3306
12       0     0 ACCEPT     6    --  *      *       10.0.2.15            0.0.0.0/0            tcp dpt:3306 state NEW,EST
ABLISHED
```

- Se muestra una regla para permitir HTTP (puerto 80). Esto permite acceder a páginas web.

```
=== OUTPUT ===
Chain OUTPUT (policy DROP 2 packets, 7568 bytes)
num   pkts bytes target     prot opt in     out     source               destination
1       16  1600 ACCEPT     0    --  *      lo      0.0.0.0/0            0.0.0.0/0
2       91  174K ACCEPT     0    --  *      *       0.0.0.0/0            0.0.0.0/0            state RELATED,ESTABLISHED
3        0     0 DROP       0    --  *      *       0.0.0.0/0            0.0.0.0/0            state INVALID
4        0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp spt:22 state ESTABLISH
ED
5        0     0 ACCEPT     1    --  *      *       0.0.0.0/0            0.0.0.0/0            icmptype 0
6        0     0 ACCEPT     1    --  *      *       0.0.0.0/0            0.0.0.0/0            icmptype 8
7        4   292 ACCEPT     17   --  *      *       0.0.0.0/0            0.0.0.0/0            udp dpt:53
8        0     0 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            tcp dpt:53
9        5   392 ACCEPT     6    --  *      *       0.0.0.0/0            0.0.0.0/0            multiport dports 80,443
10       0     0 ACCEPT     6    --  *      *       0.0.0.0/0            10.0.2.15            tcp spt:3306 state ESTABLI
SHED

=== FORWARD ===
Chain FORWARD (policy DROP 0 packets, 0 bytes)
num   pkts bytes target     prot opt in     out     source               destination
```

- Aparece una regla para permitir HTTPS (puerto 443). Sirve para navegar en páginas seguras.

```
sebastian@UbuntuPc:~$ ssh localhost
ssh: connect to host localhost port 22: Connection refused
sebastian@UbuntuPc:~$
```

- Aquí se usa la opción multiport. Esto sirve para permitir varios puertos en una sola regla (por ejemplo 80 y 443).

```
sebastian@UbuntuPc:~$ sudo systemctl status ssh
Unit ssh.service could not be found.
sebastian@UbuntuPc:~$
```

- Se muestra una regla para bloquear una dirección IP específica. La idea es impedir que esa IP acceda al servidor web.

```
sebastian@UbuntuPc:~$ sudo systemctl start ssh
```

- Se observa una prueba donde se intenta acceder desde esa IP bloqueada. La imagen demuestra que el firewall funciona correctamente.

```
sebastian@UbuntuPc:~$ sudo systemctl start ssh
sebastian@UbuntuPc:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/sshd.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
sebastian@UbuntuPc:~$ ssh localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is SHA256:4wEGeJOAaeGjgs8Oqp8P4R20g3xEkeqOMwtebsItUj4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
sebastian@localhost's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-19-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

31 updates can be applied immediately.
3 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
```

- Se ve el uso del comando nslookup. Sirve para obtener la dirección IP de la página www.josedomingo.org.

```
Expanded Security Maintenance for Applications is not enabled.

31 updates can be applied immediately.
3 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

sebastian@UbuntuPc:~$
```

- Aquí aparece una regla que bloquea esa IP obtenida con nslookup. Esto cumple uno de los requisitos del taller.

```
sebastian@UbuntuPc:~$ curl -v http://localhost
Command 'curl' not found, but can be installed with:
sudo apt install curl
sebastian@UbuntuPc:~$
```

- Se muestra la prueba del bloqueo del sitio web. El navegador o la terminal no puede acceder, lo que confirma que la regla funciona.

```
sebastian@UbuntuPc:~$ curl -v http://localhost
* Host localhost:80 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:80...
* connect to ::1 port 80 from ::1 port 53574 failed: Connection refused
*   Trying 127.0.0.1:80...
* connect to 127.0.0.1 port 80 from 127.0.0.1 port 59810 failed: Connection refused
* Failed to connect to localhost port 80 after 0 ms: Couldn't connect to server
* Closing connection
curl: (7) Failed to connect to localhost port 80 after 0 ms: Couldn't connect to server
sebastian@UbuntuPc:~$
```

- Se observa una regla para permitir acceso al servidor MySQL (puerto 3306) solo desde una IP específica.

```
sebastian@UbuntuPc:~$ curl -v http://www.josedomingo.org
* Host www.josedomingo.org:80 was resolved.
* IPv6: (none)
* IPv4: 37.187.119.60
*   Trying 37.187.119.60:80...
* Connected to www.josedomingo.org (37.187.119.60) port 80
> GET / HTTP/1.1
> Host: www.josedomingo.org
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 301 Moved Permanently
< Date: Fri, 20 Mar 2026 16:19:34 GMT
< Server: Apache/2.4.66 (Debian)
< Location: https://www.josedomingo.org/
< Content-Length: 361
< Content-Type: text/html; charset=iso-8859-1
<
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="https://www.josedomingo.org/">here</a>.</p>
<hr>
<address>Apache/2.4.66 (Debian) Server at www.josedomingo.org Port 80</address>
</body></html>
* Connection #0 to host www.josedomingo.org left intact
sebastian@UbuntuPc:~$
```

- Se hace una prueba desde la IP permitida. Aquí sí se puede acceder al servidor MySQL.

```
sebastian@UbuntuPc:~$ sudo mysql -h 10.0.2.15 -u root
[sudo] password for sebastian:
ERROR 2003 (HY000): Can't connect to MySQL server on '10.0.2.15:3306' (111)
sebastian@UbuntuPc:~$ sudo systemctl start mysql
sebastian@UbuntuPc:~$ sudo mysql -h 10.0.2.15 -u root
ERROR 2003 (HY000): Can't connect to MySQL server on '10.0.2.15:3306' (111)
sebastian@UbuntuPc:~$ sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf


sebastian@UbuntuPc:~$ sudo systemctl restart mysql
sebastian@UbuntuPc:~$ sudo ss -tlnp | grep 3306
LISTEN 0      151          0.0.0.0:3306       0.0.0.0:*    users:(("mysqld",pid=12710,fd=23))
LISTEN 0      70         127.0.0.1:33060      0.0.0.0:*    users:(("mysqld",pid=12710,fd=21))
sebastian@UbuntuPc:~$ sudo mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

- Se hace otra prueba desde una IP diferente. En esta imagen se demuestra que el acceso es bloqueado.

```
sebastian@UbuntuPc:~$ mysql -h 10.0.2.15 -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

- Se muestra el script completo en Shell con todas las reglas juntas.

```
sebastian@UbuntuPc:~$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=16.1 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=7.23 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=37.5 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2007ms
rtt min/avg/max/mdev = 7.231/20.288/37.536/12.721 ms
sebastian@UbuntuPc:~$
```

- Es el resultado final del firewall funcionando correctamente.
-

Sirve como evidencia final de que el taller se realizó completo.

EL SCRIPT DE SHELL

Acá dejamos el Shell que es el firewall.sh

https://github.com/felipe-pineros/SEGURIDAD-DE-INFORMACI-N.git